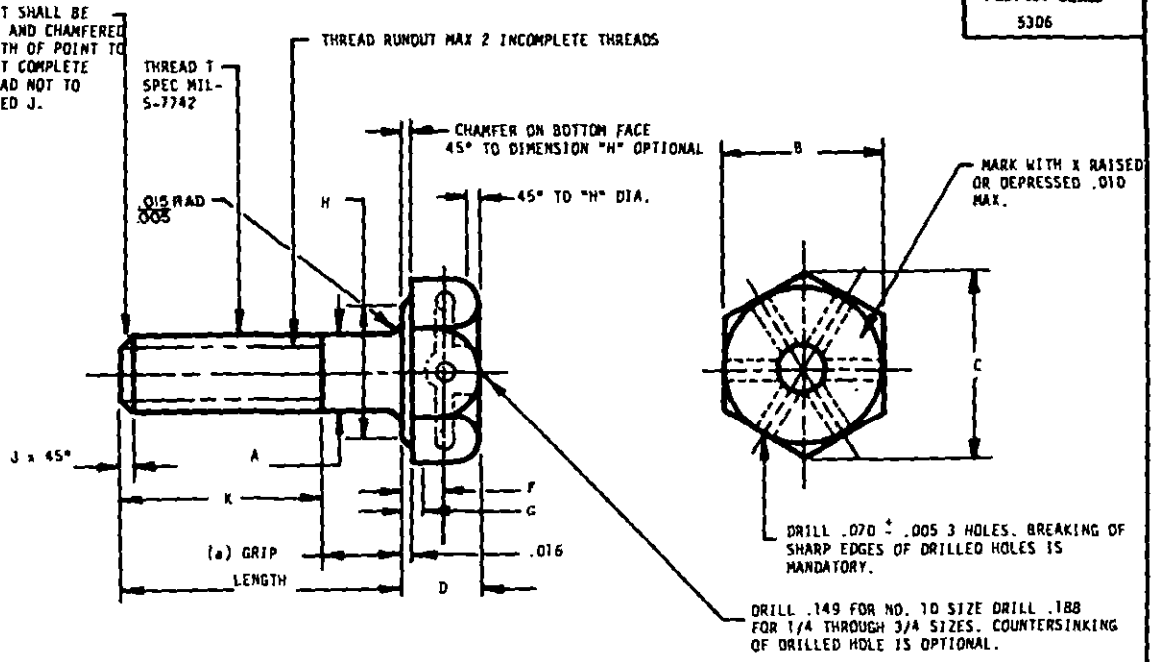
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SIZE DASH NO.	THREAD T	A DIA.		B		(b) C REF	D		F MIN	G +.000 -.010	H DIA.		J		RATED STRENGTH, LBS (c) TENSION AT ROOT DIA.	RATED STRENGTH, LBS SINGLE SHEAR AT FULL DIA.
		MAX.	MIN.	MAX.	MIN.		MAX.	MIN.			MAX.	MIN.	MAX.	MIN.		
03-	NO. 10-24UNC-3A	.189	.186	.376	.367	.430	.203	.172	.039	.120	.391	.359	.047	.015	1 800	2 125
04-	1/4-20 UNC-3A	.249	.246	.439	.430	.510	.234	.203	.052	.144	.454	.422			3 360	3 680
05-	5/16 -18 UNC-3A	.312	.309	.501	.492	.580	.297	.266	.078	.172	.516	.484			5 660	5 750
06-	3/8 -16 UNC-3A	.374	.371	.564	.554	.650					.579	.547	.063	.031	8 470	8 800
07-	7/16 -14 UNC-3A	.437	.433	.689	.679	.790	.344	.313			.704	.672			11 680	11 250
08-	1/2-13 UNC-3A	.499	.495	.751	.741	.870	.391	.359	.094	.188	.766	.734			15 730	14 700
09-	9/16 -12 UNC-3A	.562	.558	.876	.865	1.010	.438	.406	.109	.203	.891	.859			20 300	18 700
10-	5/8-11 UNC-3A	.624	.620	.939	.928	1.090	.484	.453	.140	.234	.954	.922	.078	.046	25 100	23 000
12-	3/4-10 UNC-3A	.749	.744	1.064	1.053	1.230	.578	.547	.187	.281	1.079	1.047			37 800	33 150

(a) GRIP LENGTH OF BOLTS SHALL BE MEASURED FROM THE UNDERSIDE OF THE HEAD TO END OF THE FULL CYLINDRICAL PORTION OF THE SHANK. COMPLETE THREADS SHALL BEGIN WITHIN TWO-THREAD PITCH MAXIMUM. TWO-THREAD PITCH MAXIMUM MAY CONSIST OF INCOMPLETE THREAD OR EXTRUSION ANGLE.

(b) REFERENCE DIMENSIONS ARE FOR DESIGN PURPOSES ONLY AND ARE NOT AN INSPECTION REQUIREMENT.
(c) MINIMUM YIELD STRENGTH = 76.7 PERCENT OF RATED TENSION.

MATERIAL: STEEL, SEE PROCUREMENT SPECIFICATIONS.

(B) PLATING: CADMIUM PLATE QQ-P-416, TYPE II, CLASS 2.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010 , ANGLES ± 5°.
EXAMPLES OF PART NUMBERS: MS20074-05-07 = 5/16-18 UNC-3A BOLT, .375 GRIP, .922 LONG.
UNASSIGNED DASH NUMBERS SHALL NOT BE USED.

THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.

REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

THIS DOCUMENT HAS BEEN PROMULGATED BY THE DEPARTMENT OF DEFENSE AS THE MILITARY STANDARD TO LIMIT THE SELECTION OF THE ITEM, PRODUCT, OR DESIGN COVERED HEREIN IN ENGINEERING, DESIGN, AND PROCUREMENT. THIS STANDARD SHALL BECOME EFFECTIVE NOT LATER THAN 90 DAYS AFTER THE LATEST DATE OF APPROVAL SHOWN.

APPROVED 16 SEPT 58 REVISED (A) 28 MAY 64 (B) 13 JUNE 1985

P. A.	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
AIR FORCE - 82 Other Cust ARMY - AV NAVY - AS AIR FORCE - 99		BOLT, MACHINE, AIRCRAFT, DRILLED HEAD, COARSE THREAD	MS 20074
PROCUREMENT SPECIFICATION MIL-B-6812	SUPERSEDES: AN73 THRU AN81		SHEET 1 OF 3

DD FORM 672-1 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE Project Number 5306-0960

USER ACTIVITIES:

REVIEWER ACTIVITIES: ARMY - MI, ME, DLA - IS

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LENGTH DASH NO.	-03			-04			-05			-06			-07		
	GRIP ±.016	LENGTH +.031 -.016	K REF (MIN)	GRIP ±.016	LENGTH +.031 -.016	K REF (MIN)	GRIP ±.016	LENGTH +.031 -.016	K REF (MIN)	GRIP ±.016	LENGTH +.031 -.016	K REF (MIN)	GRIP ±.016	LENGTH +.031 -.016	K REF (MIN)
02	.062	.344	.250												
03	.062	.469	.375	.062	.469	.375									
04	.062	.594	.500	.062	.594	.500	.062	.609	.515						
05	.125	.656	.499	.125	.656	.499	.125	.672	.515	.062	.734	.640			
06	.250	.781	.499	.250	.781	.499	.250	.797	.515	.125	.797	.640	.062	.797	.703
07	.375	.906	.499	.375	.906	.499	.375	.922	.515	.250	.922	.640	.188	.922	.702
10	.500	1.031	.499	.500	1.031	.499	.500	1.047	.515	.375	1.047	.640	.312	1.047	.703
11	.625	1.156	.499	.625	1.156	.499	.625	1.172	.515	.500	1.172	.640	.438	1.172	.702
12	.750	1.281	.499	.750	1.281	.499	.750	1.297	.515	.625	1.297	.640	.562	1.297	.703
13	.875	1.406	.499	.875	1.406	.499	.875	1.422	.515	.750	1.422	.640	.688	1.422	.702
14	1.000	1.531	.499	1.000	1.531	.499	1.000	1.547	.515	.875	1.547	.640	.812	1.547	.703
15	1.125	1.656	.499	1.125	1.656	.499	1.125	1.672	.515	1.000	1.672	.640	.938	1.672	.702
16	1.250	1.781	.499	1.250	1.781	.499	1.250	1.797	.515	1.125	1.797	.640	1.062	1.797	.703
17	1.375	1.906	.499	1.375	1.906	.499	1.375	1.922	.515	1.250	1.922	.640	1.188	1.922	.702
20	1.500	2.031	.499	1.500	2.031	.499	1.500	2.047	.515	1.375	2.047	.640	1.312	2.047	.703
21	1.625	2.156	.499	1.625	2.156	.499	1.625	2.172	.515	1.500	2.172	.640	1.438	2.172	.702
22	1.750	2.281	.499	1.750	2.281	.499	1.750	2.297	.515	1.625	2.297	.640	1.562	2.297	.703
23	1.875	2.406	.499	1.875	2.406	.499	1.875	2.422	.515	1.750	2.422	.640	1.688	2.422	.702
24	2.000	2.531	.499	2.000	2.531	.499	2.000	2.547	.515	1.875	2.547	.640	1.812	2.547	.703
25	2.125	2.656	.499	2.125	2.656	.499	2.125	2.672	.515	2.000	2.672	.640	1.938	2.672	.702
26	2.250	2.781	.499	2.250	2.781	.499	2.250	2.797	.515	2.125	2.797	.640	2.062	2.797	.703
27	2.375	2.906	.499	2.375	2.906	.499	2.375	2.922	.515	2.250	2.922	.640	2.188	2.922	.702
30	2.500	3.031	.499	2.500	3.031	.499	2.500	3.049	.515	2.375	3.049	.640	2.312	3.049	.703
31	2.625	3.156	.499	2.625	3.156	.499	2.625	3.172	.515	2.500	3.172	.640	2.438	3.172	.702
32	2.750	3.281	.499	2.750	3.281	.499	2.750	3.297	.515	2.625	3.297	.640	2.562	3.297	.703
33	2.875	3.406	.499	2.875	3.406	.499	2.875	3.422	.515	2.750	3.422	.640	2.688	3.422	.702
34	3.000	3.531	.499	3.000	3.531	.499	3.000	3.547	.515	2.875	3.547	.640	2.812	3.547	.703
35	3.125	3.656	.499	3.125	3.656	.499	3.125	3.672	.515	3.000	3.672	.640	2.938	3.672	.702
36	3.250	3.781	.499	3.250	3.781	.499	3.250	3.797	.515	3.125	3.797	.640	3.062	3.797	.703
37	3.375	3.906	.499	3.375	3.906	.499	3.375	3.922	.515	3.250	3.922	.640	3.188	3.922	.702
40	3.500	4.031	.499	3.500	4.031	.499	3.500	4.047	.515	3.375	4.047	.640	3.312	4.047	.703
41	3.625	4.156	.499	3.625	4.156	.499	3.625	4.172	.515	3.500	4.172	.640	3.438	4.172	.702
42	3.750	4.281	.499	3.750	4.281	.499	3.750	4.297	.515	3.625	4.297	.640	3.562	4.297	.703
43	3.875	4.406	.499	3.875	4.406	.499	3.875	4.422	.515	3.750	4.422	.640	3.688	4.422	.702
44	4.000	4.531	.499	4.000	4.531	.499	4.000	4.547	.515	3.875	4.547	.640	3.812	4.547	.703
45	4.125	4.656	.499	4.125	4.656	.499	4.125	4.672	.515	4.000	4.672	.640	3.938	4.672	.702
46	4.250	4.781	.499	4.250	4.781	.499	4.250	4.797	.515	4.125	4.797	.640	4.062	4.797	.703
47	4.375	4.906	.499	4.375	4.906	.499	4.375	4.922	.515	4.250	4.922	.640	4.188	4.922	.702
50	4.500	5.031	.499	4.500	5.031	.499	4.500	5.047	.515	4.375	5.047	.640	4.312	5.047	.703
51	4.625	5.156	.499	4.625	5.156	.499	4.625	5.172	.515	4.500	5.172	.640	4.438	5.172	.702
52	4.750	5.281	.499	4.750	5.281	.499	4.750	5.297	.515	4.625	5.297	.640	4.562	5.297	.703
53	4.875	5.406	.499	4.875	5.406	.499	4.875	5.422	.515	4.750	5.422	.640	4.688	5.422	.702
54	5.000	5.531	.499	5.000	5.531	.499	5.000	5.547	.515	4.875	5.547	.640	4.812	5.547	.703
55	5.125	5.656	.499	5.125	5.656	.499	5.125	5.672	.515	5.000	5.672	.640	4.938	5.672	.702
56	5.250	5.781	.499	5.250	5.781	.499	5.250	5.797	.515	5.125	5.797	.640	5.062	5.797	.703
57	5.375	5.906	.499	5.375	5.906	.499	5.375	5.922	.515	5.250	5.922	.640	5.188	5.922	.702
60	5.500	6.031	.499	5.500	6.031	.499	5.500	6.047	.515	5.375	6.047	.640	5.312	6.047	.703

APPROVED 18 SEP 84 REVISED (B) FOR CHANGES SEE SHEETS 1 AND 2

USER ACTIVITIES:

REVIEWER ACTIVITIES:
ARMY - MI
DLA - IS

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AIR FORCE 82 Other Cust ARMY AV NAVY AS AIR FORCE 99	INTERNATIONAL INTEREST	TITLE BOLT, MACHINE, AIRCRAFT, DRILLED HEAD, COARSE THREAD	MILITARY STANDARD MS 20074
PROCUREMENT SPECIFICATION MIL-B-6812	SUPERSEDES		SHEET 2 OF 3

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LENGTH DASH NO.	-08			-09			-10			-12		
	GRIP ±.016	LENGTH +.031 -.016	K REF (MIN)	GRIP ±.016	LENGTH +.031 -.016	K REF (MIN)	GRIP ±.016	LENGTH +.031 -.016	K REF (MIN)	GRIP ±.016	LENGTH +.031 -.016	K REF (MIN)
10	.250	1.047	.765									
11	.375	1.172	.765									
12	.500	1.297	.765	.438	1.312	.842						
13	.625	1.422	.765	.562	1.438	.844						
14	.750	1.547	.765	.688	1.562	.842	.562	1.562	.968			
15	.875	1.672	.765	.812	1.688	.844	.688	1.688	.968			
16	1.000	1.797	.765	.938	1.812	.842	.812	1.812	.968	.750	1.812	1.030
17	1.125	1.922	.765	1.062	1.938	.844	.938	1.938	.968	.875	1.938	1.031
20	1.250	2.047	.765	1.188	2.062	.842	1.062	2.062	.968	1.000	2.062	1.030
21	1.375	2.172	.765	1.312	2.188	.844	1.188	2.188	.968	1.125	2.188	1.031
22	1.500	2.297	.765	1.438	2.312	.842	1.312	2.312	.968	1.250	2.312	1.030
23	1.625	2.422	.765	1.562	2.438	.844	1.438	2.438	.968	1.375	2.438	1.031
24	1.750	2.547	.765	1.688	2.562	.842	1.562	2.562	.968	1.500	2.562	1.030
25	1.875	2.672	.765	1.812	2.688	.844	1.688	2.688	.968	1.625	2.688	1.031
26	2.000	2.797	.765	1.938	2.812	.842	1.812	2.812	.968	1.750	2.812	1.030
27	2.125	2.922	.765	2.062	2.938	.844	1.938	2.938	.968	1.875	2.938	1.031
30	2.250	3.047	.765	2.188	3.062	.842	2.062	3.062	.968	2.000	3.062	1.030
31	2.375	3.172	.765	2.312	3.188	.844	2.188	3.188	.968	2.125	3.188	1.031
32	2.500	3.297	.765	2.438	3.312	.842	2.312	3.312	.968	2.250	3.312	1.030
33	2.625	3.422	.765	2.562	3.438	.844	2.438	3.438	.968	2.375	3.438	1.031
34	2.750	3.547	.765	2.688	3.562	.842	2.562	3.562	.968	2.500	3.562	1.030
35	2.875	3.672	.765	2.812	3.688	.844	2.688	3.688	.968	2.625	3.688	1.031
36	3.000	3.797	.765	2.938	3.812	.842	2.812	3.812	.968	2.750	3.812	1.030
37	3.125	3.922	.765	3.062	3.938	.844	2.938	3.938	.968	2.875	3.938	1.031
40	3.250	4.047	.765	3.188	4.062	.842	3.062	4.062	.968	3.000	4.062	1.030
41	3.375	4.172	.765	3.312	4.188	.844	3.188	4.188	.968	3.125	4.188	1.031
42	3.500	4.297	.765	3.438	4.312	.842	3.312	4.312	.968	3.250	4.312	1.030
43	3.625	4.422	.765	3.562	4.438	.844	3.438	4.438	.968	3.375	4.438	1.031
44	3.750	4.547	.765	3.688	4.562	.842	3.562	4.562	.968	3.500	4.562	1.030
45	3.875	4.672	.765	3.812	4.688	.844	3.688	4.688	.968	3.625	4.688	1.031
46	4.000	4.797	.765	3.938	4.812	.842	3.812	4.812	.968	3.750	4.812	1.030
47	4.125	4.922	.765	4.062	4.938	.844	3.938	4.938	.968	3.875	4.938	1.031
50	4.250	5.047	.765	4.188	5.062	.842	4.062	5.062	.968	4.000	5.062	1.030
51	4.375	5.172	.765	4.312	5.188	.844	4.188	5.188	.968	4.125	5.188	1.031
52	4.500	5.297	.765	4.438	5.312	.842	4.312	5.312	.968	4.250	5.312	1.030
53	4.625	5.422	.765	4.562	5.438	.844	4.438	5.438	.968	4.375	5.438	1.031
54	4.750	5.547	.765	4.688	5.562	.842	4.562	5.562	.968	4.500	5.562	1.030
55	4.875	5.672	.765	4.812	5.688	.844	4.688	5.688	.968	4.625	5.688	1.031
56	5.000	5.797	.765	4.938	5.812	.842	4.812	5.812	.968	4.750	5.812	1.030
57	5.125	5.922	.765	5.062	5.938	.844	4.938	5.938	.968	4.875	5.938	1.031
60	5.250	6.047	.765	5.188	6.016	.842	5.062	6.062	.968	5.000	6.062	1.030

APPROVED 18 SEP 64 REVISED (B) FOR CHANGES SEE SHEETS 1 AND 2

USER ACTIVITIES:

REVIEWER ACTIVITIES:
ARMY - MI, ME
DLA - 15

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P.A. AIR FORCE 82 Other Cost ARMY- AV NAVY- AS AIR FORCE 99	INTERNATIONAL INTEREST	TITLE: BOLT, MACHINE, AIRCRAFT, DRILLED HEAD, COARSE THREAD	MILITARY STANDARD MS20074
PROCUREMENT SPECIFICATION MIL-B-6812	SUPERSEDES:		SHEET 3 OF 3

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